

Trainings ▾

Preparatory Steps

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

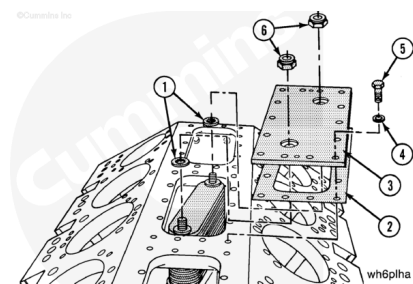
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove the turbochargers. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Remove the turbochargers. Refer to Procedure 010-037 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-037-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-008 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-008-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html)
- Remove the turbochargers. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr-gc.html)
- Remove the exhaust manifolds. Refer to Procedure 011-008 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-008-tr-gc.html)

Remove

The K38 and QSK38 engines have three oil cooler covers (older K38 engines have two large cover plates) and the K50 and QSK50 engines have four oil cooler covers. All covers are removed the same way. Each cover contains two threaded holes that can be used to aid in the removal of the cover.

Remove the two locknuts (6). Remove all the capscrews. Use two of the capscrews in the threaded holes to remove the plate (3). Remove the oil cooler cover. Remove and discard the gaskets (1 and 2).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the covers with solvent and dry with compressed air.

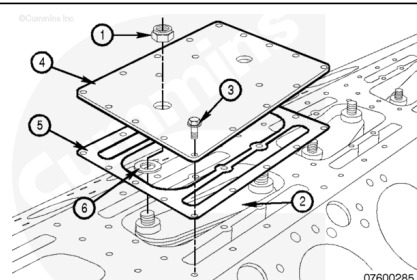
Inspect the covers for corrosion and other damage.

Replace the covers if corroded or otherwise damaged.

Install

The K38 and QSK38 engines have three oil cooler covers and the K50 and QSK50 engines have four oil cooler covers.

Note : Older K38 engines have **only two large cover plates**. The older covers have plain copper washers. Newer covers have steel lock washers.



Install the two gaskets (1) on each oil cooler element. Install the cover gasket (2) on the cylinder block.

Install the cover plates (3), lock washer (4), and capscrews (5).

Install the two lock nuts (6) for each oil cooler element.

Tighten the oil cooler cover capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

Tighten the self-locking nuts (21 and 22).

Torque Value:

Initial Torque Value 45 n•m [33 ft-lb]

Torque Value:

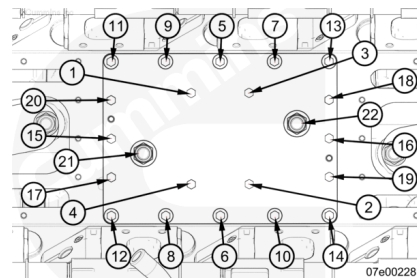
Final Torque Value 95 n•m [70 ft-lb]

For industrial engines, tighten the two-stage frame support bracket capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

For four turbocharger engines, tighten the exhaust collector support bracket capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the exhaust manifolds. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
- Remove the exhaust manifolds. Refer to Procedure 011-008 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-008-tr.html)
- Install the turbochargers. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Install the turbochargers. Refer to Procedure 010-037 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-037-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html).
- Operate the engine to normal operating temperature. Check for leaks.

- Install the exhaust manifolds. Refer to Procedure 011-008 in Section 11.
(/qs3/pubsys2/xml/en/procedures/28/28-011-008-tr-gc.html)
- Install the turbochargers. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr-gc.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html)
- Start the engine. Check for leaks.

Last Modified: 13-May-2019
